

Train Test Split

```
In [1]: import os
```

```
import pandas as pd
```

```
ROOT_PATH = os.path.dirname(os.getcwd())
```

```
DATA_FOLDER_PATH = os.path.join(ROOT_PATH, 'data', 'processed')
```

```
In [2]: df = pd.read_parquet(os.path.join(DATA_FOLDER_PATH, 'cleaned_arrivals.parquet'))
```

```
df.head()
```

```
Out[2]:
```

	access_route	state	origin_country	month	year	arrival_count	month_num	date	y
0	Aérea	Bahia	Afeganistão	Janeiro	2026	1.0	01	2026-01-31	
1	Aérea	São Paulo	Afeganistão	Janeiro	2026	4.0	01	2026-01-31	
2	Terrestre	Paraná	Afeganistão	Janeiro	2026	1.0	01	2026-01-31	
3	Terrestre	Rio Grande do Sul	Afeganistão	Janeiro	2026	1.0	01	2026-01-31	
4	Marítima	Rio de Janeiro	Afeganistão	Janeiro	2026	3.0	01	2026-01-31	

Split by Time

The test set is always the **last TEST_MONTHS = 12 months** of the data, and everything before it is training data. The split date is **derived from the data**, never typed by hand: when a new CSV is added and the notebooks are re-run, the split rolls forward on its own.

Every later notebook reads its dates from `train.parquet` and `test.parquet`, so this cell is the single place that defines the forecasting horizon.

Two checks guard the split:

- **No missing months.** The aggregated monthly series must be contiguous from the first to the last month. A gap would shift the 12-month window.
- **Last month complete.** The ministry sometimes publishes partial months. If the last month is below 50% of the same month one year earlier, it is probably incomplete, and a warning is printed. Remove that month's rows (or wait for the full release) before modelling.

```

In [3]: TEST_MONTHS = 12
INCOMPLETE_MONTH_RATIO = 0.5

monthly_totals = df.groupby('date')['arrival_count'].sum().sort_index()

expected_months = pd.date_range(
    monthly_totals.index.min(), monthly_totals.index.max(), freq='ME'
)
missing_months = expected_months.difference(monthly_totals.index)
assert missing_months.empty, f'Missing months in the data: {list(missing_months)}'

last_month = monthly_totals.index.max()
test_start = last_month - pd.offsets.MonthEnd(TEST_MONTHS - 1)

print(
    f'Complete time range: {monthly_totals.index.min():%B %Y} to {last_month:%B %Y}'
    f'({len(monthly_totals)} months)'
)
print(f'Test window      : {test_start:%B %Y} to {last_month:%B %Y}')

same_month_last_year = last_month - pd.offsets.MonthEnd(12)
ratio = monthly_totals[last_month] / monthly_totals[same_month_last_year]
print(
    f'Last month vs the same month one year earlier: {ratio:.2f} '
    f'({monthly_totals[last_month]:,.0f} vs '
    f'{monthly_totals[same_month_last_year]:,.0f})'
)
if ratio < INCOMPLETE_MONTH_RATIO:
    print(
        f'WARNING: {last_month:%B %Y} looks incomplete. Remove it from the raw CSV'
        'or wait for the full release before modelling.'
    )

```

Complete time range: January 1989 to August 2026 (452 months)
 Test window : September 2025 to August 2026
 Last month vs the same month one year earlier: 1.00 (576,276 vs 576,049)

```

In [4]: train_df = df[df['date'] < test_start]
print(
    f'Train set time range: {train_df["date"].min():%d %B %Y} to '
    f'{train_df["date"].max():%d %B %Y}'
)
train_df.tail()

```

Train set time range: 31 January 1989 to 31 August 2025

Out[4]:	access_route	state	origin_country	month	year	arrival_count	month_num	d
	976740	Aérea	Paraná	Índia	Agosto	2025	1.0	08 2008
	976741	Aérea	Distrito Federal	Índia	Agosto	2025	11.0	08 2008
	976742	Terrestre	Paraná	Índia	Agosto	2025	1.0	08 2008
	976743	Terrestre	Paraná	Índia	Agosto	2025	31.0	08 2008
	976744	Terrestre	Rio Grande do Sul	Índia	Agosto	2025	2.0	08 2008

```
In [5]: test_df = df[df['date'] >= test_start]
assert test_df['date'].nunique() == TEST_MONTHS
print(
    f'Test set time range: {test_df["date"].min():%d %B %Y} to '
    f'{test_df["date"].max():%d %B %Y}'
)
test_df.tail()
```

Test set time range: 30 September 2025 to 31 August 2026

Out[5]:	access_route	state	origin_country	month	year	arrival_count	month_num
	985269	Terrestre	Rio Grande do Sul	Índia	Dezembro	2025	5.0 12
	985270	Marítima	Alagoas	Índia	Dezembro	2025	3.0 12
	985271	Marítima	Santa Catarina	Índia	Dezembro	2025	3.0 12
	985272	Marítima	Santa Catarina	Índia	Dezembro	2025	5.0 12
	985273	Fluvial	Amapá	Índia	Dezembro	2025	2.0 12

Output

```
In [6]: train_df.to_parquet(os.path.join(DATA_FOLDER_PATH, 'train.parquet'), index=False)
test_df.to_parquet(os.path.join(DATA_FOLDER_PATH, 'test.parquet'), index=False)
```